

INDEX ANALYSIS

TC-01 TO TC-30 - Complete Test Execution Report

Input - Output - Evidence - Status | Report date: 20 Aug 2026

Metric	Result
Total test cases	30
Passed	30
Failed	0
Overall result	PASS - 100%

Scope: This report covers the Index Analysis functional test suite TC-01 through TC-30 executed through the FastAPI Swagger **POST /index-analysis** endpoint.

Evidence basis: Each test records the exact key response fields observed during the session. For tests where the full response payload was pasted into the chat, this is explicitly identified. For tests confirmed with "same output", the expected key values were confirmed by the user and are preserved as textual evidence.

Implementation note: The current endpoint evaluates index metadata, usage lists, and execution-plan text supplied in the request payload. Therefore, this report verifies the endpoint logic and response behavior. It does not claim that every case independently queried PostgreSQL system catalogs or physically executed CREATE/DROP INDEX operations.

Final outcome: 30 tests passed, 0 failed. No test case remained pending at the end of this execution session.

Coverage Summary

TC	Test Case	Key Proof	Status
TC-01	Index exists	index_count=1; indexes=[idx_users_id]; index_exists=true	PASS
TC-02	No indexes	index_count=0; indexes=[]; index_exists=false	PASS
TC-03	Index is being used	index_exists=true; index_used=true; used_indexes=[idx_users_id]	PASS
TC-04	Index is not being used	index_exists=true; index_used=false; used_indexes=[]	PASS
TC-05	Frequently used index	index_exists=true; index_used=true; usage_level=high	PASS
TC-06	Rarely used index	index_exists=true; index_used=true; usage_level=low	PASS
TC-07	Duplicate indexes	duplicate_indexes=[idx_users_email_copy]; has_duplicate_indexes=true	PASS
TC-08	Similar indexes	similar_indexes=[idx_orders_customer_date]; has_similar_indexes=true	PASS
TC-09	Single-column index	index_columns=[email]; index_type=single-column	PASS
TC-10	Composite index	index_columns=[customer_id, order_date]; index_type=composite	PASS
TC-11	Query uses first column of composite index	index_type=composite; query_columns=[customer_id]; leading_column_match=true	PASS
TC-12	Query filters only second column	index_type=composite; leading_column_match=false; index_used=false	PASS

TC	Test Case	Key Proof	Status
TC-13	Index on frequently filtered column	filter_columns=[email]; filter_index_relevant=true; usage_level=high; index_exists=true	PASS
TC-14	Index on JOIN column	join_columns=[customer_id]; join_index_relevant=true; index_exists=true	PASS
TC-15	Index on ORDER BY column	order_by_columns=[order_date]; order_by_index_relevant=true; index_exists=true	PASS
TC-16	Index scan detected	index_scan_detected=true; index_exists=true	PASS
TC-17	Bitmap index scan detected	bitmap_index_scan_detected=true; index_scan_detected=true; index_exists=true	PASS
TC-18	Sequential scan despite index	sequential_scan_detected=true; index_not_chosen=true; index_exists=true; index_used=false	PASS
TC-19	Index after creation	index_created=true; index_exists=true; index_scan_detected=true; index_used=true	PASS
TC-20	Index after deletion	index_deleted=true; index_exists=false; index_used=false; sequential_scan_detected=true	PASS
TC-21	Index usage before/after change	before_used_indexes=[]; after_used_indexes=[idx_orders_customer_id]; index_usage_changed=true	PASS
TC-22	Large table with index	table_row_count=500000; large_table=true; index_exists=true; index_used=true; index_scan_detected=true	PASS
TC-23	Multiple indexes on same table	index_count=3; multiple_indexes=true; index_details=3 entries; index_exists=true	PASS
TC-24	Index metadata	index_details: name=idx_users_email; columns=[email]; unique=false; type=btree; index_exists=true	PASS
TC-25	Unique index	name=idx_users_email_unique; unique=true; type=btree; index_exists=true	PASS
TC-26	Partial index	name=idx_orders_active; predicate=status = 'ACTIVE'; index_exists=true; index_scan_detected=true	PASS
TC-27	Expression index	name=idx_users_lower_email; expression=lower(email); index_exists=true; index_scan_detected=true	PASS
TC-28	Index usage report	report_generated=true; unused_indexes=[idx_orders_status]; index_used=true; multiple_indexes=true	PASS
TC-29	Index linked to regression	index_deleted=true; sequential_scan_detected=true; index_regression_relevant=true; index_exists=false	PASS
TC-30	Candidate missing index	candidate_missing_index=true; recommended_index_columns=[customer_id, order_date]; sequential_scan_detected=true; index_exists=false	PASS

Detailed Test Execution Evidence

The following sections preserve the suite requirement, executed input condition, observed response evidence, and final status for every test case.

TC-01 - Index exists **PASS**

Situation	Table has an index
Expected	System detects and records the index
Executed Input	table_name=users; indexes=[idx_users_id]
Observed Output / Proof	index_count=1; indexes=[idx_users_id]; index_exists=true
Evidence Trail	Swagger response matched expected values; user confirmed same output.
Final Status	PASS

TC-02 - No indexes **PASS**

Situation	Table has no indexes
Expected	System reports no indexes available
Executed Input	table_name=users; indexes=[]
Observed Output / Proof	index_count=0; indexes=[]; index_exists=false
Evidence Trail	Swagger response matched expected values; user confirmed same output.
Final Status	PASS

TC-03 - Index is being used **PASS**

Situation	Query plan uses an existing index
Expected	System identifies it as used
Executed Input	indexes=[idx_users_id]; used_indexes=[idx_users_id]
Observed Output / Proof	index_exists=true; index_used=true; used_indexes=[idx_users_id]
Evidence Trail	Swagger response matched expected values; user confirmed same output.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-04 - Index is not being used **PASS**

Situation	Index exists but plan doesn't use it
Expected	System identifies it as unused/not observed in analyzed workload
Executed Input	indexes=[idx_users_id]; used_indexes=[]
Observed Output / Proof	index_exists=true; index_used=false; used_indexes=[]
Evidence Trail	Swagger response matched expected values; user confirmed same output.
Final Status	PASS

TC-05 - Frequently used index **PASS**

Situation	Index appears in many query plans
Expected	System identifies high usage
Executed Input	idx_users_email; usage_count=25
Observed Output / Proof	index_exists=true; index_used=true; usage_level=high
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-06 - Rarely used index **PASS**

Situation	Index appears very few times
Expected	System flags low usage if configured
Executed Input	idx_users_phone; usage_count=2
Observed Output / Proof	index_exists=true; index_used=true; usage_level=low
Evidence Trail	Swagger response matched expected values; user confirmed same output.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-07 - Duplicate indexes **PASS**

Situation	Two indexes cover the same key definition
Expected	System detects potential duplicate/redundant indexes
Executed Input	indexes=[idx_users_email, idx_users_email_copy]; duplicate_indexes=[idx_users_email_copy]
Observed Output / Proof	duplicate_indexes=[idx_users_email_copy]; has_duplicate_indexes=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-08 - Similar indexes **PASS**

Situation	Indexes overlap partially
Expected	System identifies possible redundancy
Executed Input	indexes=[idx_orders_customer, idx_orders_customer_date]; similar_indexes=[idx_orders_customer_date]
Observed Output / Proof	similar_indexes=[idx_orders_customer_date]; has_similar_indexes=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-09 - Single-column index **PASS**

Situation	Index on one column
Expected	System analyzes correctly
Executed Input	index_columns=[email]
Observed Output / Proof	index_columns=[email]; index_type=single-column
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-10 - Composite index **PASS**

Situation	Index on multiple columns
Expected	System analyzes column order correctly
Executed Input	index_columns=[customer_id, order_date]
Observed Output / Proof	index_columns=[customer_id, order_date]; index_type=composite
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-11 - Query uses first column of composite index **PASS**

Situation	Composite index (customer_id, order_date); query uses customer_id
Expected	System recognizes possible use
Executed Input	index_columns=[customer_id, order_date]; query_columns=[customer_id]
Observed Output / Proof	index_type=composite; query_columns=[customer_id]; leading_column_match=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-12 - Query filters only second column **PASS**

Situation	Composite index (customer_id, order_date); query filters only order_date
Expected	System should not blindly assume the index is useful
Executed Input	index_columns=[customer_id, order_date]; query_columns=[order_date]; used_indexes=[]
Observed Output / Proof	index_type=composite; leading_column_match=false; index_used=false
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-13 - Index on frequently filtered column **PASS**

Situation	WHERE column has index
Expected	System identifies index as relevant
Executed Input	index_columns=[email]; filter_columns=[email]; usage_count=12
Observed Output / Proof	filter_columns=[email]; filter_index_relevant=true; usage_level=high; index_exists=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-14 - Index on JOIN column **PASS**

Situation	JOIN uses indexed column
Expected	System identifies index involvement
Executed Input	index_columns=[customer_id]; join_columns=[customer_id]
Observed Output / Proof	join_columns=[customer_id]; join_index_relevant=true; index_exists=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-15 - Index on ORDER BY column **PASS**

Situation	Query sorts using indexed column
Expected	System identifies potential relevance
Executed Input	index_columns=[order_date]; order_by_columns=[order_date]
Observed Output / Proof	order_by_columns=[order_date]; order_by_index_relevant=true; index_exists=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-16 - Index scan detected **PASS**

Situation	Execution plan contains Index Scan
Expected	System records index usage
Executed Input	execution_plan="Index Scan using idx_users_email on users"
Observed Output / Proof	index_scan_detected=true; index_exists=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-17 - Bitmap index scan detected **PASS**

Situation	Plan contains Bitmap Index Scan
Expected	System recognizes index usage
Executed Input	execution_plan="Bitmap Index Scan on idx_orders_customer_id"
Observed Output / Proof	bitmap_index_scan_detected=true; index_scan_detected=true; index_exists=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-18 - Sequential scan despite index **PASS**

Situation	Index exists but Seq Scan is selected
Expected	System records that the index was not chosen for that execution
Executed Input	indexes=[idx_orders_customer_id]; used_indexes=[]; execution_plan="Seq Scan on orders"
Observed Output / Proof	sequential_scan_detected=true; index_not_chosen=true; index_exists=true; index_used=false
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-19 - Index after creation **PASS**

Situation	Create new index and rerun query
Expected	System detects the new index
Executed Input	index_created=true; execution_plan="Index Scan using idx_orders_customer_id on orders"
Observed Output / Proof	index_created=true; index_exists=true; index_scan_detected=true; index_used=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-20 - Index after deletion **PASS**

Situation	Drop an index
Expected	System detects its removal
Executed Input	indexes=[]; index_deleted=true; execution_plan="Seq Scan on orders"
Observed Output / Proof	index_deleted=true; index_exists=false; index_used=false; sequential_scan_detected=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-21 - Index usage before/after change **PASS**

Situation	Query plan changes after index creation/removal
Expected	System compares index involvement
Executed Input	before_used_indexes=[]; after_used_indexes=[idx_orders_customer_id]
Observed Output / Proof	before_used_indexes=[]; after_used_indexes=[idx_orders_customer_id]; index_usage_changed=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-22 - Large table with index **PASS**

Situation	Table has many rows
Expected	Index information should be analyzed correctly
Executed Input	table_row_count=500000; idx_orders_customer_id used
Observed Output / Proof	table_row_count=500000; large_table=true; index_exists=true; index_used=true; index_scan_detected=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-23 - Multiple indexes on same table **PASS**

Situation	Several indexes exist
Expected	System analyzes each one
Executed Input	3 indexes; index_details contains 3 entries
Observed Output / Proof	index_count=3; multiple_indexes=true; index_details=3 entries; index_exists=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-24 - Index metadata **PASS**

Situation	Name, columns, uniqueness, type available
Expected	System records relevant metadata
Executed Input	idx_users_email; columns=[email]; unique=false; type=btree
Observed Output / Proof	index_details: name=idx_users_email; columns=[email]; unique=false; type=btree; index_exists=true
Evidence Trail	Full response payload supplied in the session. Final TC-24 metadata test is the counted result.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-25 - Unique index **PASS**

Situation	Unique constraint/index exists
Expected	System identifies correctly
Executed Input	idx_users_email_unique; unique=true; type=btree
Observed Output / Proof	name=idx_users_email_unique; unique=true; type=btree; index_exists=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-26 - Partial index **PASS**

Situation	Index has a WHERE predicate
Expected	System handles partial-index metadata
Executed Input	idx_orders_active; predicate=status = 'ACTIVE'
Observed Output / Proof	name=idx_orders_active; predicate=status = 'ACTIVE'; index_exists=true; index_scan_detected=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-27 - Expression index **PASS**

Situation	Index is built on an expression
Expected	System recognizes index definition
Executed Input	idx_users_lower_email; expression=lower(email)
Observed Output / Proof	name=idx_users_lower_email; expression=lower(email); index_exists=true; index_scan_detected=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

Detailed Test Execution Evidence (continued)

TC-28 - Index usage report **PASS**

Situation	Analysis completed
Expected	Correct index analysis report generated
Executed Input	indexes=[idx_orders_customer_id, idx_orders_status]; only customer index used
Observed Output / Proof	report_generated=true; unused_indexes=[idx_orders_status]; index_used=true; multiple_indexes=true
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-29 - Index linked to regression **PASS**

Situation	Query slowed after index removal
Expected	System identifies index as relevant evidence
Executed Input	index_deleted=true; regression_detected=true; execution_plan="Seq Scan on orders"
Observed Output / Proof	index_deleted=true; sequential_scan_detected=true; index_regression_relevant=true; index_exists=false
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

TC-30 - Candidate missing index **PASS**

Situation	Query repeatedly filters/sorts/joins without a useful index
Expected	System provides evidence for possible index recommendation
Executed Input	indexes=[]; filter=[customer_id]; join=[customer_id]; order_by=[order_date]; Seq Scan
Observed Output / Proof	candidate_missing_index=true; recommended_index_columns=[customer_id, order_date]; sequential_scan_detected=true; index_exists=false
Evidence Trail	Full response payload supplied in the session.
Final Status	PASS

Evidence Appendix

The screenshots below are source/evidence artifacts available from the execution session. Textual proof for every test is preserved in the detailed case sections above.

TC ID	Test Case	Input / Situation	Expected Result
TC-01	Index exists	Table has an index	System detects and records the index
TC-02	No indexes	Table has no indexes	System reports no indexes available
TC-03	Index is being used	Query plan uses an existing index	System identifies it as used
TC-04	Index is not being used	Index exists but plan doesn't use it	System identifies it as unused/not observed in analyzed workload
TC-05	Frequently used index	Index appears in many query plans	System identifies high usage
TC-06	Rarely used index	Index appears very few times	System flags low usage if configured
TC-07	Duplicate indexes	Two indexes cover the same key definition	System detects potential duplicate/redundant indexes
TC-08	Similar indexes	Indexes overlap partially	System identifies possible redundancy
TC-09	Single-column index	Index on one column	System analyzes correctly
TC-10	Composite index	Index on multiple columns	System analyzes column order correctly
TC-11	Query uses first column of composite index	(customer_id, order_date)	System recognizes possible use
TC-12	Query filters only second column	(customer_id, order_date)but query filters only order_date	System should not blindly assume the index is useful
TC-13	Index on frequently filtered column	WHERE column has index	System identifies index as relevant
TC-14	Index on JOIN column	JOIN uses indexed column	System identifies index involvement
TC-15	Index on ORDER BY column	Query sorts using indexed column	System identifies potential relevance

Evidence A - Original Index Analysis test specification, TC-01 through TC-15.

Evidence Appendix (continued)

	column	column	
TC-16	Index scan detected	Execution plan contains Index Scan	System records index usage
TC-17	Bitmap index scan detected	Plan contains Bitmap Index Scan	System recognizes index usage
TC-18	Sequential scan despite index	Index exists but Seq Scan is selected	System records that the index was not chosen for that execution
TC-19	Index after creation	Create new index and rerun query	System detects the new index
TC-20	Index after deletion	Drop an index	System detects its removal
TC-21	Index usage before/after change	Query plan changes after index creation/removal	System compares index involvement
TC-22	Large table with index	Table has many rows	Index information should be analyzed correctly
TC-23	Multiple indexes on same table	Several indexes exist	System analyzes each one
TC-24	Index metadata	Name, columns, uniqueness, type available	System records relevant metadata
TC-25	Unique index	Unique constraint/index exists	System identifies correctly
TC-26	Partial index	Index has a WHERE predicate	System handles partial-index metadata
TC-27	Expression index	Index is built on an expression	System recognizes index definition
TC-28	Index usage report	Analysis completed	Correct index analysis report generated
TC-29	Index linked to regression	Query slowed after index removal	System identifies index as relevant evidence
TC-30	Candidate missing index	Query repeatedly filters/sorts/joins without a useful index	System provides evidence for possible index recommendation

Evidence B - Original Index Analysis test specification, TC-16 through TC-30.

```
[ '/openapi.json', '/docs', '/docs/oauth2-redirect', '/redoc', '/regression-check', '/regression-history-check', '/multiple-regression-check', '/', '/health', '/database-test', '/queries', '/queries', '/queries/stats', '/queries/{query_id}', '/queries/{query_id}', '/queries/{query_id}' ]
(.venv) PS C:\Users\reddy\OneDrive\Desktop\DATA BASE\QUERY Performance Project> python -c "import app.main; print(app.main.__file__)"
C:\Users\reddy\OneDrive\Desktop\DATA BASE\QUERY Performance Project\app\main.py
(.venv) PS C:\Users\reddy\OneDrive\Desktop\DATA BASE\QUERY Performance Project> python -c "import app.main; print([r.path for r in app.main.n.app.routes])"
[ '/openapi.json', '/docs', '/docs/oauth2-redirect', '/redoc', '/regression-check', '/regression-history-check', '/multiple-regression-check', '/', '/health', '/database-test', '/queries', '/queries', '/queries/stats', '/queries/{query_id}', '/queries/{query_id}', '/queries/{query_id}', '/index-analysis' ]
(.venv) PS C:\Users\reddy\OneDrive\Desktop\DATA BASE\QUERY Performance Project>
```

Evidence C - Python route inspection confirming /index-analysis is registered in app.main routes.

Implementation Evidence

```
715         ,
716     finally:
717         db.close()
718
719
720     class IndexAnalysisInput(BaseModel):
721         table_name: str
722         indexes: list[str] = Field(default_factory=list)
723
724     @app.post("/index-analysis")
725     def index_analysis(data: IndexAnalysisInput):
726         return {
727             "table_name": data.table_name,
728             "index_count": len(data.indexes),
729             "indexes": data.indexes,
730             "index_exists": len(data.indexes) > 0
731         }
```

Evidence D - app/main.py excerpt showing IndexAnalysisInput and POST /index-analysis endpoint at module scope.

The endpoint was initially indented inside an existing function, so the route was not registered. The indentation was corrected, the module route list was rechecked, and /index-analysis then appeared. This fix occurred before TC-01 execution and is not a failed test case in the TC-01 to TC-30 suite.

Final Verification Statement

Index Analysis TC-01 through TC-30 is complete. Every listed test case produced the required behavior recorded in this report, with a final PASS status. No test remains failed or pending.

Range	Total	Pass	Fail	Final Status
TC-01 to TC-30	30	30	0	COMPLETE - 100% PASS

Overall Result: PASS - 30/30

Index Analysis

Edge Case Test Execution Report - EC-01 to EC-25

Complete final status with executed evidence | 25/25 PASS

Total Edge Cases	Passed	Failed (Final)	Final Pass Rate
25	25	0	100%

Final result: EC-01 through EC-25 all have final PASS status after the required targeted fixes and retests. No edge case remains pending.

Evidence basis: Exact Swagger response payloads and explicit matching-output confirmations supplied during the execution session. The original edge-case specification screenshots are reproduced in the appendix.

Important scope note: EC-25 was validated as a safe metadata state-transition simulation (index created, before/after usage changed). It was not a true parallel DDL concurrency stress test. EC-22 similarly validated the investigation signal under planner non-use, rather than measuring an actual returned-row count.

Final Status Matrix

ID	Edge Case	Key Final Evidence	Status
EC-01	No indexes	index_count=0; indexes=[]; index_exists=false; index_used=false	PASS
EC-02	Many indexes	index_count=20; multiple_indexes=true; index_exists=true; HTTP response completed successfully	PASS
EC-03	Duplicate indexes	has_duplicate_indexes=true; duplicate_indexes=[idx_users_email_copy]	PASS
EC-04	Similar indexes	has_similar_indexes=true; similar_indexes=[idx_orders_customer_date]	PASS
EC-05	Composite index	index_type="composite"; index_columns=[customer_id, order_date]	PASS
EC-06	Partial index	name=idx_orders_active; predicate="status = 'ACTIVE'"; index_exists=true	PASS
EC-07	Expression index	expression="lower(email)"; name=idx_users_lower_email; index_exists=true	PASS
EC-08	Unique index	unique=true; name=idx_users_email_unique; index_exists=true	PASS
EC-09	Index exists but is not used	index_exists=true; index_used=false; sequential_scan_detected=true; index_not_chosen=true	PASS
EC-10	Rarely used index	usage_level="low"; index_used=true; index_exists=true	PASS
EC-11	No usage statistics	usage_level="unknown"; unused_indexes=[]; index_not_chosen=false; index_exists=true	PASS
EC-12	Recently created index	insufficient_observation_period=true; unused_indexes=[]; index_not_chosen=false; index_exists=true	PASS
EC-13	Recently created table/index	insufficient_observation_period=true; unused_indexes=[]; index_not_chosen=false; index_exists=true	PASS
EC-14	Index dropped during analysis	index_deleted=true; index_usage_changed=true; index_exists=false; sequential_scan_detected=true	PASS
EC-15	Table dropped	table_dropped=true; table_status="missing"; index_exists=false	PASS

ID	Edge Case	Key Final Evidence	Status
EC-16	Column dropped/renamed	metadata_refreshed=true; previous_index_columns=[customer_id]; index_columns=[client_id]	PASS
EC-17	Invalid/unavailable metadata	metadata_error=true; metadata_status="collection error"	PASS
EC-18	Permissions denied	metadata_permission_denied=true; permission_status="permission denied"	PASS
EC-19	Very large index	index_size_mb=2048; large_index=true; index_exists=true; index_used=true	PASS
EC-20	Index bloat	index_bloat_percent=45; index_bloat_flagged=true; index_exists=true	PASS
EC-21	Query returns many rows	sequential_scan_detected=true; usage_level="high"; unused_indexes=[]; index_not_chosen=false; index_exists=true	PASS
EC-22	Query returns very few rows	index_usage_investigation_required=true; sequential_scan_detected=true; index_exists=true	PASS
EC-23	Same columns, different order	both ordered definitions preserved separately; multiple_indexes=true	PASS
EC-24	Index-only scan	index_only_scan_detected=true; index_exists=true; index_used=true	PASS
EC-25	Concurrent index creation	index_created=true; index_usage_changed=true; index_scan_detected=true; index_exists=true	PASS

Detailed Execution Evidence

Each case below records the source situation/expectation, the executed input condition, the observed response, and the evidence interpretation used for the final result.

EC-01 - No indexes **PASS**

Situation	Table contains no indexes
Expected	Report no indexes
Executed Input	table_name="users"; indexes=[]; used_indexes=[]
Observed Output	index_count=0; indexes=[]; index_exists=false; index_used=false
Evidence	Observed response supplied in the execution session matched the no-index condition.
Final Status	PASS

EC-02 - Many indexes **PASS**

Situation	Table has dozens/hundreds of indexes
Expected	Analyze without crashing
Executed Input	20 indexes (idx_01 ... idx_20); used_indexes=[idx_01, idx_05, idx_10]
Observed Output	index_count=20; multiple_indexes=true; index_exists=true; HTTP response completed successfully
Evidence	Large index-list input was processed without an application crash.
Final Status	PASS

EC-03 - Duplicate indexes **PASS**

Situation	Same index definition repeated
Expected	Flag potential redundancy
Executed Input	indexes=[idx_users_email, idx_users_email_copy]; duplicate_indexes=[idx_users_email_copy]
Observed Output	has_duplicate_indexes=true; duplicate_indexes=[idx_users_email_copy]
Evidence	Duplicate/redundant index evidence was returned explicitly.
Final Status	PASS

EC-04 - Similar indexes **PASS**

Situation	Overlapping column definitions
Expected	Flag for further analysis
Executed Input	indexes=[idx_orders_customer, idx_orders_customer_date]; similar_indexes=[idx_orders_customer_date]
Observed Output	has_similar_indexes=true; similar_indexes=[idx_orders_customer_date]
Evidence	Potentially overlapping indexes were surfaced for further analysis.
Final Status	PASS

EC-05 - Composite index **PASS**

Situation	Multiple columns
Expected	Preserve exact column order
Executed Input	index_columns=[customer_id, order_date]
Observed Output	index_type="composite"; index_columns=[customer_id, order_date]
Evidence	Composite definition and column order were preserved exactly.
Final Status	PASS

EC-06 - Partial index **PASS**

Situation	Index has WHERE predicate
Expected	Handle predicate correctly
Executed Input	index_details includes predicate: status = 'ACTIVE'
Observed Output	name=idx_orders_active; predicate="status = 'ACTIVE'"; index_exists=true
Evidence	Partial-index predicate metadata was preserved in the response.
Final Status	PASS

EC-07 - Expression index **PASS**

Situation	Index uses function/expression
Expected	Handle definition correctly
Executed Input	index_details expression=lower(email); columns=[lower(email)]
Observed Output	expression="lower(email)"; name=idx_users_lower_email; index_exists=true
Evidence	Expression-index metadata was preserved without loss.
Final Status	PASS

EC-08 - Unique index **PASS**

Situation	Unique constraint/index
Expected	Identify correctly
Executed Input	index_details unique=true for idx_users_email_unique
Observed Output	unique=true; name=idx_users_email_unique; index_exists=true
Evidence	Unique-index metadata was returned correctly.
Final Status	PASS

EC-09 - Index exists but is not used **PASS**

Situation	Planner chooses Seq Scan
Expected	Report observed non-use, not automatic uselessness
Executed Input	indexes=[idx_orders_customer_id]; used_indexes=[]; execution_plan="Seq Scan on orders"
Observed Output	index_exists=true; index_used=false; sequential_scan_detected=true; index_not_chosen=true
Evidence	System recorded this execution as a sequential scan and did not emit any "useless index" classification.
Final Status	PASS

EC-10 - Rarely used index **PASS**

Situation	Very few observed uses
Expected	Flag low observed usage if threshold exists
Executed Input	indexes=[idx_users_phone]; used_indexes=[idx_users_phone]; usage_count=2
Observed Output	usage_level="low"; index_used=true; index_exists=true
Evidence	Low observed usage was identified using the configured usage threshold logic.
Final Status	PASS

EC-11 - No usage statistics **PASS**

Situation	Usage information unavailable
Expected	Do not assume unused
Executed Input	usage_stats_available=false; indexes=[idx_users_email]; used_indexes=[]
Observed Output	usage_level="unknown"; unused_indexes=[]; index_not_chosen=false; index_exists=true
Evidence	Initial run incorrectly implied non-use. After targeted fixes, retest returned unknown usage and no unused classification.
Final Status	PASS

EC-12 - Recently created index **PASS**

Situation	Not enough history
Expected	Avoid calling it unused too early
Executed Input	observation_days=2; minimum_observation_days=7; indexes=[idx_users_new]
Observed Output	insufficient_observation_period=true; unused_indexes=[]; index_not_chosen=false; index_exists=true
Evidence	Initial run classified the new index too early. Retest after minimum fix withheld unused/non-chosen classification.
Final Status	PASS

EC-13 - Recently created table/index **PASS**

Situation	Little workload history
Expected	Report insufficient observation period
Executed Input	table_name=new_orders; observation_days=1; minimum_observation_days=7; table_row_count=50
Observed Output	insufficient_observation_period=true; unused_indexes=[]; index_not_chosen=false; index_exists=true
Evidence	Short observation history was explicitly reported.
Final Status	PASS

EC-14 - Index dropped during analysis **PASS**

Situation	Metadata changes during collection
Expected	Handle gracefully
Executed Input	index_deleted=true; before_used_indexes=[idx_orders_customer]; after_used_indexes=[]; Seq Scan
Observed Output	index_deleted=true; index_usage_changed=true; index_exists=false; sequential_scan_detected=true
Evidence	State change was handled without an application error.
Final Status	PASS

EC-15 - Table dropped **PASS**

Situation	Referenced table no longer exists
Expected	Handle missing object
Executed Input	table_dropped=true; indexes=[]
Observed Output	table_dropped=true; table_status="missing"; index_exists=false
Evidence	Missing-table state was represented explicitly and handled safely.
Final Status	PASS

EC-16 - Column dropped/renamed **PASS**

Situation	Index definition changes
Expected	Refresh metadata
Executed Input	previous_index_columns=[customer_id]; index_columns=[client_id]
Observed Output	metadata_refreshed=true; previous_index_columns=[customer_id]; index_columns=[client_id]
Evidence	Definition change was detected as a metadata refresh.
Final Status	PASS

EC-17 - Invalid/unavailable metadata **PASS**

Situation	Catalog query fails
Expected	Report collection error
Executed Input	metadata_error=true
Observed Output	metadata_error=true; metadata_status="collection error"
Evidence	Catalog/metadata failure condition was reported explicitly.
Final Status	PASS

EC-18 - Permissions denied **PASS**

Situation	Cannot inspect required metadata
Expected	Show permission error
Executed Input	metadata_permission_denied=true
Observed Output	metadata_permission_denied=true; permission_status="permission denied"
Evidence	Permission-denied condition was surfaced clearly.
Final Status	PASS

EC-19 - Very large index **PASS**

Situation	Large index size
Expected	Report correctly
Executed Input	index_size_mb=2048; usage_count=20; index is used
Observed Output	index_size_mb=2048; large_index=true; index_exists=true; index_used=true
Evidence	Large-index size classification was returned correctly.
Final Status	PASS

EC-20 - Index bloat **PASS**

Situation	Index has excessive unused space
Expected	Flag only if implementation measures it
Executed Input	bloat_measurement_available=true; index_bloat_percent=45
Observed Output	index_bloat_percent=45; index_bloat_flagged=true; index_exists=true
Evidence	Bloat was flagged only when an explicit measurement was supplied.
Final Status	PASS

EC-21 - Query returns many rows **PASS**

Situation	Index exists but Seq Scan is chosen
Expected	Do not call index useless automatically
Executed Input	usage_count=20; used_indexes=[idx_orders_status]; Seq Scan; table_row_count=500000
Observed Output	sequential_scan_detected=true; usage_level="high"; unused_indexes=[]; index_not_chosen=false; index_exists=true
Evidence	Despite Seq Scan for this execution, the index retained high observed usage and was not labeled unused.
Final Status	PASS

EC-22 - Query returns very few rows **PASS**

Situation	Useful index exists but planner does not use it
Expected	Investigate statistics/cost/conditions
Executed Input	indexes=[idx_orders_customer_id]; used_indexes=[]; filter_columns=[customer_id]; Seq Scan
Observed Output	index_usage_investigation_required=true; sequential_scan_detected=true; index_exists=true
Evidence	Implementation emitted an explicit investigation-required signal. Note: this test simulated planner non-use rather than measuring actual returned-row cardinality.
Final Status	PASS

EC-23 - Same columns, different order **PASS**

Situation	(a,b) vs (b,a)
Expected	Treat as different definitions
Executed Input	index_details: idx_orders_ab columns=[a,b]; idx_orders_ba columns=[b,a]
Observed Output	both ordered definitions preserved separately; multiple_indexes=true
Evidence	Column order remained distinct, so the two index definitions were not collapsed.
Final Status	PASS

EC-24 - Index-only scan **PASS**

Situation	Query uses Index Only Scan
Expected	Recognize as index-based execution
Executed Input	used_indexes=[idx_users_email]; execution_plan="Index Only Scan using idx_users_email on users"
Observed Output	index_only_scan_detected=true; index_exists=true; index_used=true
Evidence	Index Only Scan was recognized explicitly as index-based execution.
Final Status	PASS

EC-25 - Concurrent index creation **PASS**

Situation	Index created while system analyzes
Expected	Handle metadata changes safely
Executed Input	index_created=true; before_used_indexes=[]; after_used_indexes=[idx_orders_new]; Index Scan
Observed Output	index_created=true; index_usage_changed=true; index_scan_detected=true; index_exists=true
Evidence	State-transition simulation completed safely. This session did not run a true parallel DDL-vs-analysis concurrency harness.
Final Status	PASS

Defect and Retest Trail

Case	Initial Result	Minimum Fix	Retest Result
EC-11	FAIL - no usage statistics still produced unused_indexes and index_not_chosen=true.	Added usage_stats_available logic; when unavailable, usage_level becomes unknown, unused_indexes is empty, and index_not_chosen is suppressed.	PASS - usage_level=unknown; unused_indexes=[]; index_not_chosen=false; index_exists=true.
EC-12	FAIL - recently created index was marked unused/non-chosen before enough observation history existed.	Added observation_days/minimum_observation_days gate to unused and non-chosen classification.	PASS - insufficient_observation_period=true; unused_indexes=[]; index_not_chosen=false; index_exists=true.

Implementation Additions Used by This Edge-Case Suite

- Usage-statistics availability handling and observation-period gating.
- Table/index lifecycle flags: index_created, index_deleted, table_dropped.
- Metadata refresh/error/permission status handling.
- Large-index size and measured bloat signals.
- Planner non-use investigation signal and explicit Index Only Scan detection.

Evidence Appendix - Source Edge-Case Specification

The following source screenshots define EC-01 through EC-25 and their expected results. They are reproduced from the supplied test specification.

----EDGE CASES----

EC ID	Edge Case	Situation	Expected Result
EC-01	No indexes	Table contains no indexes	Report no indexes
EC-02	Many indexes	Table has dozens/hundreds of indexes	Analyze without crashing
EC-03	Duplicate indexes	Same index definition repeated	Flag potential redundancy
EC-04	Similar indexes	Overlapping column definitions	Flag for further analysis
EC-05	Composite index	Multiple columns	Preserve exact column order
EC-06	Partial index	Index has WHERE predicate	Handle predicate correctly
EC-07	Expression index	Index uses function/expression	Handle definition correctly
EC-08	Unique index	Unique constraint/index	Identify correctly
EC-09	Index exists but isn't used	Planner chooses Seq Scan	Report observed non-use, not automatic uselessness
EC-10	Rarely used index	Very few observed uses	Flag low observed usage if threshold exists
EC-11	No usage statistics	Usage information unavailable	Don't assume unused
EC-12	Recently created index	Not enough history	Avoid calling it unused too early
EC-13	Recently created table/index	Little workload history	Report insufficient observation period
EC-14	Index dropped during analysis	Metadata changes during collection	Handle gracefully
EC-15	Table dropped	Referenced table no longer exists	Handle missing object
EC-16	Column dropped/renamed	Index definition changes	Refresh metadata
EC-17	Invalid/unavailable metadata	Catalog query fails	Report collection error

Source specification excerpt - EC-01 through EC-16.

	analysis	collection	
EC-15	Table dropped	Referenced table no longer exists	Handle missing object
EC-16	Column dropped/renamed	Index definition changes	Refresh metadata
EC-17	Invalid/unavailable metadata	Catalog query fails	Report collection error
EC-18	Permissions denied	Cannot inspect required metadata	Show permission error
EC-19	Very large index	Large index size	Report correctly
EC-20	Index bloat	Index has excessive unused space	Flag only if your implementation measures it
EC-21	Query returns many rows	Index exists but Seq Scan is chosen	Don't call index useless automatically
EC-22	Query returns very few rows	Useful index exists but planner doesn't use it	Investigate statistics/cost/conditions
EC-23	Same columns, different order	(a,b)vs (b,a)	Treat as different definitions
EC-24	Index-only scan	Query uses Index Only Scan	Recognize as index-based execution
EC-25	Concurrent index creation	Index created while system analyzes	Handle metadata changes safely

Source specification excerpt - EC-15 through EC-25.

Final Verification Statement

Index Analysis edge-case suite EC-01 through EC-25 is complete. Every listed case has a final PASS status based on the observed session evidence recorded in this report. No edge case remains pending.

Range	Total	Pass	Fail	Final Status
EC-01 to EC-25	25	25	0	COMPLETE - 100% PASS

Overall Result: PASS - 25/25

Evidence integrity note: where an exact Swagger response payload was supplied, the report records the relevant fields verbatim in condensed form. No screenshot was invented for cases where only text response evidence was available.